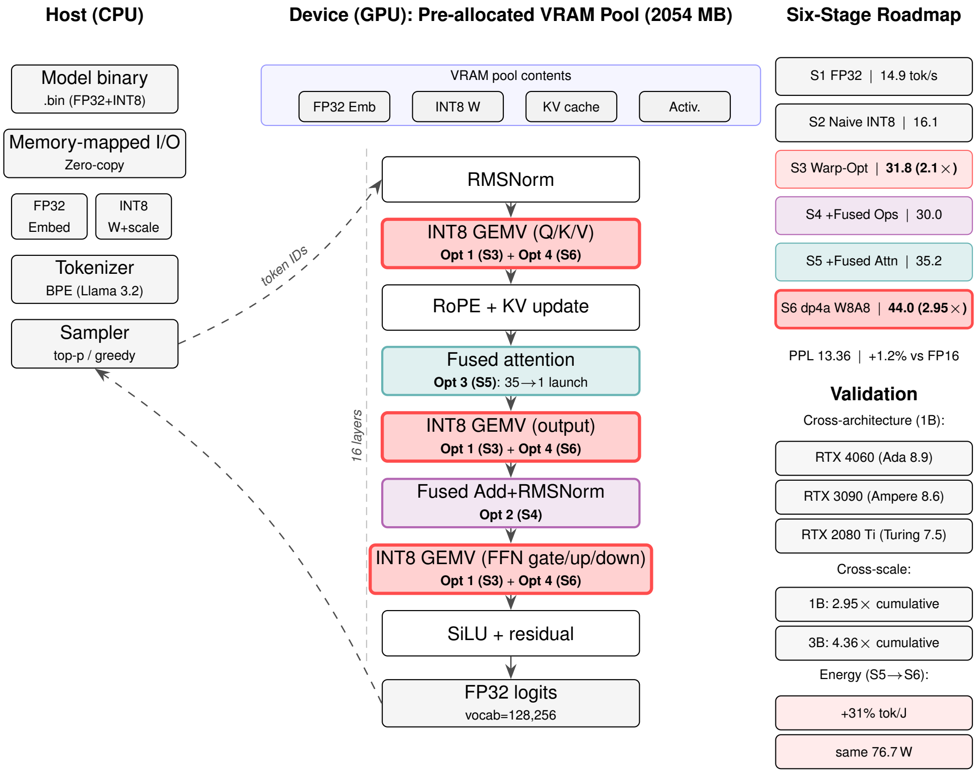
Host (CPU)
Device (GPU): Pre-allocated VRAM Pool (2054 MB)
Six-Stage Roadmap
Model binary
.bin (FP32+INT8)
Memory-mapped I/O
Zero-copy
FP32
Embed
INT8
W+scale
Tokenizer
BPE (Llama 3.2)
Sampler
top-p / greedy
token IDs
VRAM pool contents
FP32 Emb
INT8 W
KV cache
Activ.
16 layers
RMSNorm
INT8 GEMV (Q/K/V)
Opt 1 (S3) + Opt 4 (S6)
RoPE + KV update
Fused attention
Opt 3 (S5): 35→1 launch
INT8 GEMV (output)
Opt 1 (S3) + Opt 4 (S6)
Fused Add+RMSNorm
Opt 2 (S4)
INT8 GEMV (FFN gate/up/down)
Opt 1 (S3) + Opt 4 (S6)
SiLU + residual
FP32 logits
vocab=128,256
S1 FP32 | 14.9 tok/s
S2 Naive INT8 | 16.1
S3 Warp-Opt | 31.8 (2.1×)
S4 +Fused Ops | 30.0
S5 +Fused Attn | 35.2
S6 dp4a W8A8 | 44.0 (2.95×)
PPL 13.36 | +1.2% vs FP16
Validation
Cross-architecture (1B):
RTX 4060 (Ada 8.9)
RTX 3090 (Ampere 8.6)
RTX 2080 Ti (Turing 7.5)
Cross-scale:
1B: 2.95× cumulative
3B: 4.36× cumulative
Energy (S5→S6):
+31% tok/J
same 76.7 W